SF2940: Exam HT23

1.a) $\bar{X}_n \xrightarrow[n\to\infty]{2} \bar{X}$, if $\mathbb{E}[(\bar{X}_n - \bar{X})^2] \xrightarrow[n\to\infty]{} 0$. //

1.b) For any $\varepsilon > 0$, we have by Markov/Tschebyscheff inequality that

$$\mathbb{P}(|\bar{X}_n - \bar{X}| > \varepsilon) \leq \frac{\mathbb{E}[(\bar{X}_n - \bar{X})^2]}{\varepsilon^2}.$$

Since $\varepsilon > 0$ and $\bar{X}_n \xrightarrow[n\to\infty]{2} \bar{X}$, we conclude that

$$\lim_{n\to\infty} \mathbb{P}(|\bar{X}_n - \bar{X}| > \varepsilon) = 0. \quad //$$

1.c.) If $\bar{X} \in \mathrm{Exp}(1)$, then $f_{\bar{X}}(x) = e^{-x} \mathbb{1}_{\{x>0\}}$.

Then, $\psi_{\bar{X}}(t) = \mathbb{E}[e^{tX}] = \int_0^\infty e^{tx} e^{-x} dx$

$$= \int_0^\infty e^{-(1-t)x} dx$$

$$= \frac{1}{1-t}$$

for all $|t| < 1$. Thus $h = 1$. //

1d.) The k'th moment of $\bar{X}$ is given by

$$\mathbb{E}[\bar{X}^k] = \psi_{\bar{X}}^{(k)}(t)\Big|_{t=0}.$$

Now with 1c) we have

$$\psi'_{\bar{X}}(t) = \frac{1}{(1-t)^2} \qquad (|t|<1)$$

$$\psi''_{\bar{X}}(t) = \frac{2}{(1-t)^3}$$

$$\psi'''_{\bar{X}}(t) = \frac{2\cdot 3}{(1-t)^4}$$

$$\vdots$$

$$\psi^{(k)}_{\bar{X}}(t) = \frac{1\cdot 2\cdot 3\cdots k}{(1-t)^{k+1}} = \frac{k!}{(1-t)^{k+1}}.$$

Setting $t=0$ above, we find $\underline{\mathbb{E}[\bar{X}^k] = k!}$

Alternatively, for $|t|<1$, $\psi_{\bar{X}}(t) = \mathbb{E}[e^{t\bar{X}}] = 1 + \sum_{k=1}^{\infty} \frac{t^k}{k!}\mathbb{E}[\bar{X}^k]$. (Taylor expansion)

On the other hand, $\psi_{\bar{X}}(t) = \frac{1}{1-t} = 1 + \sum_{k=1}^{\infty} 1\cdot t^k$. (geometric series)

Comparing coefficients, we must have $\mathbb{E}[\bar{X}^k] = k!$ //

2a.) We have by definition

$$f_{\bar{X}|\bar{Y}=y}(x) = \frac{f_{\bar{X},\bar{Y}}(x,y)}{f_{\bar{Y}}(y)}.$$

The marginal density $f_{\bar{Y}}(y)$ is given, for $y > 0$, by

$$f_{\bar{Y}}(y) = \int_{\mathbb{R}} f_{\bar{X},\bar{Y}}(x,y)\,dx = 4\int_0^y e^{-2y}\,dx$$

$$= 4ye^{-2y}.$$

Thus,

$$f_{\bar{X}|\bar{Y}=y}(x) = \frac{4e^{-2y}\mathbb{1}_{\{0<x<y\}}}{4ye^{-2y}} = \frac{1}{y}\mathbb{1}_{\{0<x<y\}}.$$

Therefore $\bar{X}|\bar{Y}=y$ is uniformly distributed on the interval $(0,y)$, and we have $\mathbb{E}[\bar{X}|\bar{Y}] = \frac{\bar{Y}}{2}$.

26.) We set $U := Y$ and $V := \frac{X}{Y-X}$.

The density $f_{X,Y}(x,y)$ is supported on $S := \{(x,y) \in \mathbb{R}^2 : 0 < x < y\}$.

Let $g_1(x,y) := y$

$g_2(x,y) := \frac{x}{y-x}$.

Then, $\vec{g} : S \longrightarrow T$ with $T := \{(u,v) \in \mathbb{R}^2 : 0 < u, 0 < v\}$.

$(x,y) \longmapsto (u,v)$

The inverse $\vec{h}$ of $\vec{g}$ is given by $(\vec{h} = (h_1, h_2))$

$\vec{h} : T \longrightarrow S$

$(u,v) \longmapsto \left(\frac{uv}{v+1}, u\right)$.

$\left(h_1(u,v) = \frac{uv}{v+1},\ h_2(u,v) = u \right)$

The Jacobian of the transformation $\vec{h}$ is

$$J = \det \begin{pmatrix} \frac{\partial h_1(u,v)}{\partial u} & \frac{\partial h_1(u,v)}{\partial v} \\ \frac{\partial h_2(u,v)}{\partial u} & \frac{\partial h_2(u,v)}{\partial v} \end{pmatrix} = \det \begin{pmatrix} \frac{v}{v+1} & \frac{u}{(v+1)^2} \\ 1 & 0 \end{pmatrix}$$

$$= \frac{-u}{(v+1)^2}.$$

By the transformation theorem,

$$f_{\bar{U},\bar{V}}(u,v) = \begin{cases} f_{\bar{X},\bar{Y}}\left(\frac{uv}{v+1}, u\right)|J|, & \text{for } (u,v)\in T, \\ & \text{(i.e. } u>0, v>0) \\ 0, & \text{else.} \end{cases}$$

$$= \begin{cases} 4e^{-2u}\frac{u}{(1+v)^2}\mathbb{1}\{0<\frac{uv}{v+1}<u\}, & \text{for } u>0, v>0, \\ 0, & \text{else.} \end{cases}$$

$$= 4e^{-2u}\frac{u}{(1+v)^2}\mathbb{1}\{0<\frac{uv}{v+1}<u, u>0, v>0\}.$$

Since if $u>0$ and $v>0$, we have that $0<\frac{uv}{v+1}<u$ is always true (because then $0<\frac{v}{1+v}<1$), we have,

$$f_{\bar{U},\bar{V}}(u,v) = 4e^{-2u}\frac{u}{(1+v)^2}\mathbb{1}\{u>0, v>0\}$$

$$= 4e^{-2u}u\,\mathbb{1}\{u>0\}\cdot\frac{1}{(1+v)^2}\mathbb{1}\{v>0\}.$$

Remark: The joint density $f_{\bar{U},\bar{V}}(u,v)$ factorizes into the product of the marginals $f_{\bar{U}}(u)$ and $f_{\bar{V}}(v)$, hence $\bar{U}$ and $\bar{V}$ are independent. In fact, $U\in\Gamma(2,\frac{1}{2})$ and $V\in F(2,2)$.

3a.) We have $\vec{Y} = B\vec{X}$ with $B = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & -1 \end{pmatrix}$,

Thus by linear transformation:

$$\Lambda_{\vec{Y}} = B \Lambda_{\vec{X}} B^T$$

$$= \begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & -1 \end{pmatrix} \begin{pmatrix} 2 & 1 & 1 \\ 1 & 3 & -1 \\ 1 & -1 & 2 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 0 \\ 1 & -1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & -1 \end{pmatrix} \begin{pmatrix} 4 & 1 \\ 3 & 2 \\ 2 & -1 \end{pmatrix}$$

$$= \begin{pmatrix} 9 & 2 \\ 2 & 2 \end{pmatrix} = \Lambda_{\vec{Y}}.$$

Hence $\vec{Y} \in \mathcal{N}\left(0, \begin{pmatrix} 9 & 2 \\ 2 & 2 \end{pmatrix}\right)$, since also $\mathbb{E}Y_1 = \mathbb{E}Y_2 = 0$.

3b.) $$f_{Y_1 | Y_2 = 0}(y_1) = \frac{f_{Y_1, Y_2}(y_1, 0)}{f_{Y_2}(0)}.$$

Since $Y_2 \in \mathcal{N}(0, 2)$, we have $f_{Y_2}(0) = \frac{1}{\sqrt{4\pi}}$.

To determine $f_{Y_1, Y_2}(y_1, y_2)$, we invert $\Lambda_{\vec{Y}}$: By Cramer's rule

$$\Lambda_{\vec{Y}}^{-1} = \frac{1}{\det \Lambda_{\vec{Y}}} \begin{pmatrix} 2 & -2 \\ -2 & 9 \end{pmatrix} = \frac{1}{14} \begin{pmatrix} 2 & -2 \\ -2 & 9 \end{pmatrix}.$$

Thus, $f_{\bar{Y}_1,\bar{Y}_2}(y_1,y_2) = \frac{1}{2\pi\sqrt{14}} \cdot e^{-\frac{1}{2}(y_1,y_2)\Lambda_Y^{-1}\begin{pmatrix}y_1\\y_2\end{pmatrix}}$.

Choosing $y_2 = 0$, we get

$$f_{\bar{Y}_1,\bar{Y}_2}(y_1,0) = \frac{1}{2\pi\sqrt{14}} e^{-\frac{1}{14}y_1^2}.$$

Thus

$$f_{\bar{Y}_1|\bar{Y}_2=0}(y_1) = \frac{\frac{1}{2\pi\sqrt{14}} e^{-\frac{1}{14}y_1^2}}{\frac{1}{\sqrt{4\pi}}} = \frac{1}{\sqrt{14\pi}} e^{-\frac{1}{14}y_1^2}.$$

3c.) Using the hint, we have

$$\max\{\bar{X}_1,\bar{X}_3\} = \frac{\bar{X}_1+\bar{X}_3}{2} + \frac{|\bar{X}_1-\bar{X}_3|}{2}.$$

Taking expectation,

$$\mathbb{E}\max\{\bar{X}_1,\bar{X}_3\} = \underbrace{\mathbb{E}\bar{X}_1}_{=0} + \underbrace{\mathbb{E}\bar{X}_3}_{=0} + \frac{\mathbb{E}|\bar{X}_1-\bar{X}_3|}{2}.$$

since $\vec{X}^2$ is centered

$$= \frac{\mathbb{E}|\bar{X}_1-\bar{X}_3|}{2} = \frac{\mathbb{E}[|\bar{Y}_2|]}{2}.$$

Since $\vec{Y} \in \mathcal{N}(0, \begin{pmatrix}9&2\\2&2\end{pmatrix})$, we have $Y_2 \in \mathcal{N}(0,2)$.

Thus, $$\mathbb{E}|\bar{Y}_2| = \frac{1}{\sqrt{4\pi}}\int |y_2| e^{-\frac{1}{4}y_2^2} dy_2$$

Symmetry $$= \frac{2}{\sqrt{4\pi}}\int_0^{\infty} y_2 e^{-\frac{1}{4}y_2^2} dy_2$$

$$= \frac{-2}{\sqrt{4\pi}}\int_0^{\infty} 2\frac{\partial}{\partial y_2} e^{-\frac{1}{4}y_2^2} dy_2$$

$$= \frac{-4}{\sqrt{4\pi}} e^{-\frac{1}{4}y_2^2}\Big|_0^{\infty} = \frac{-4}{\sqrt{4\pi}}(0-1)$$

$$= \frac{2}{\sqrt{\pi}}.$$

Hence, $\mathbb{E}[\max\{\bar{X}_1, \bar{X}_5\}] = \frac{1}{\sqrt{\pi}}$. //

4a.) First, we note that

$$\{M_n > y\} = \{\bar{X}_1 > y, \bar{X}_2 > y, \cdots, \bar{X}_n > y\}.$$

By independence, we have

$$\mathbb{P}(M_n > y) = \mathbb{P}(\bar{X}_1 > y, \bar{X}_2 > y, \cdots, \bar{X}_n > y)$$

$$= \prod_{i=1}^{n} \mathbb{P}(\bar{X}_i > y)$$

$$= \left(\mathbb{P}(\bar{X}_1 > y)\right)^n,$$

where we used that $(\bar{X}_i)$ are identically distributed.

For $y \in (0,1)$, we have

$$\mathbb{P}(\bar{X}_1 > y) = 1 - \mathbb{P}(\bar{X}_1 \le y) = 1 - 2\int_0^y (1-x)\,dx$$

$$= 1 - 2\left(y - \frac{y^2}{2}\right)$$

$= 1 - 2y + y^2$

$= (1-y)^2$, for $y \in (0,1)$.

(and $\mathbb{P}(\bar{X}_1 > y) = 0$ if $y \geq 1$; $\mathbb{P}(\bar{X}_1 > y) = 1$ if $y \in (-\infty, 0)$)

Since $M_n > 0$ we have for any $\varepsilon \in (0,1)$,

$$\mathbb{P}(|M_n - 0| > \varepsilon) = \mathbb{P}(M_n > \varepsilon) = \left(\mathbb{P}(\bar{X}_1 > \varepsilon)\right)^n$$

$$= (1-\varepsilon)^{2n}$$

$$\longrightarrow 0, \text{ for any } \varepsilon \in (0,1).$$

as $n \to \infty$

(for $\varepsilon \geq 1$ we have $\mathbb{P}(|M_n| > \varepsilon) = 0$).

Thus $M_n \xrightarrow[n\to\infty]{P} 0$ as claimed. //

4b.) We have $\mathbb{P}(n \cdot M_n \leq y) = \mathbb{P}\left(M_n \leq \frac{y}{n}\right) = 1 - \mathbb{P}\left(M_n > \frac{y}{n}\right)$

$$= 1 - \left(\mathbb{P}\left(\bar{X}_1 > \frac{y}{n}\right)\right)^n$$

$$= 1 - \left(1 - \frac{y}{n}\right)^{2n}.$$

(for any fixed $y \geq 0$, for n sufficiently large).

By the Euler formula for the exponential function,

we have $\lim_{n\to\infty} \left(1 - \frac{y}{n}\right)^{2n} = e^{-2y}$.

Thus, $F_{Y_n}(y) \xrightarrow[n\to\infty]{} F_Y(y) = \begin{cases} 1 - e^{-2y}, & y \geq 0 \\ 0, & y < 0. \end{cases}$ (the limiting CDF is continuous everywhere.)

with $Y \in Exp\left(\frac{1}{2}\right)$.

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4c.) Let $H_n := \{N = n\}$ be the event that $N = n$, $n = 1, 2, 5, \ldots$

Note that $\bigcup_{n \geq 1}^{\infty} H_n = \Omega$ and $H_n \cap H_m = \phi$; $n \neq m$.

Let $A := \{M_N > y\}$, then by the law of total probability / σ-additivity

$$\begin{aligned}
\mathbb{P}(A) &= \mathbb{P}\left(\bigcup_{n \geq 1}^{\infty} (A \cap H_n)\right) \\
&= \sum_{n=1}^{\infty} \mathbb{P}(A \cap H_n) \\
&= \sum_{n=1}^{\infty} \mathbb{P}(A \mid H_n) \cdot \mathbb{P}(H_n) \\
&= \sum_{n=1}^{\infty} \mathbb{P}(M_n > y)\, \mathbb{P}(N = n).
\end{aligned}$$

← since $(A \cap H_n) \cap (A \cap H_m) = \phi$ for $n \neq m$.

← definition of conditional probability

← since N is independent from (X_i)

By 4a) we have $\mathbb{P}(M_n > y) = \left(\mathbb{P}(\bar{X}_1 > y)\right)^n$ thus

$$\begin{aligned}
\mathbb{P}(M_N > y) &= \sum_{n=1}^{\infty} \left(\mathbb{P}(\bar{X}_1 > y)\right)^n \mathbb{P}(N = n) \\
&= g_N\left(\mathbb{P}(\bar{X}_1 > y)\right),
\end{aligned}$$

where $g_N(t) = \sum_{n=0}^{\infty} t^n \mathbb{P}(N = n)$ is the probability generating function of the r.v. N.

(note that $\mathbb{P}(N = 2) = 0$.)

Thus,

$$\mathbb{P}(M_N \leq y) = 1 - g_N\left(\mathbb{P}(\bar{X}_1 > y)\right) \text{ as claimed.}$$

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